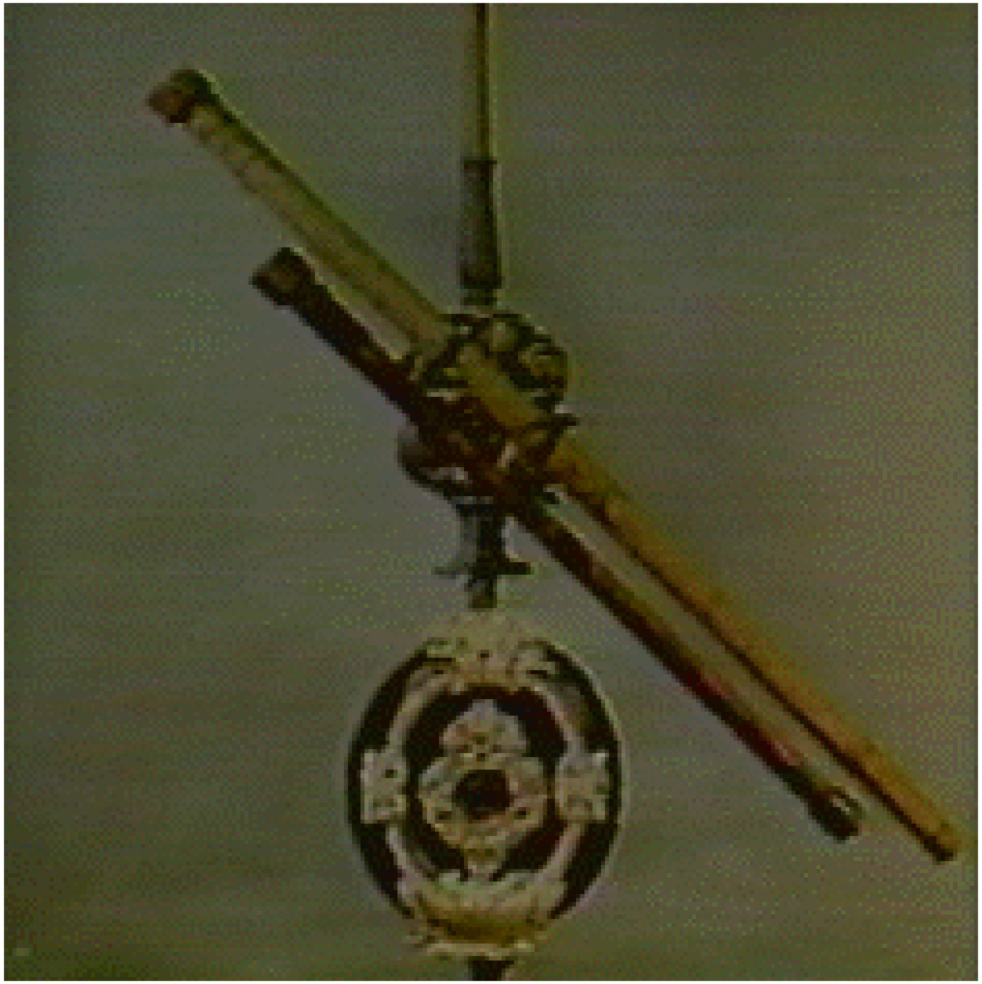
Space Instruments
– Introduction –

Week 1, part 1

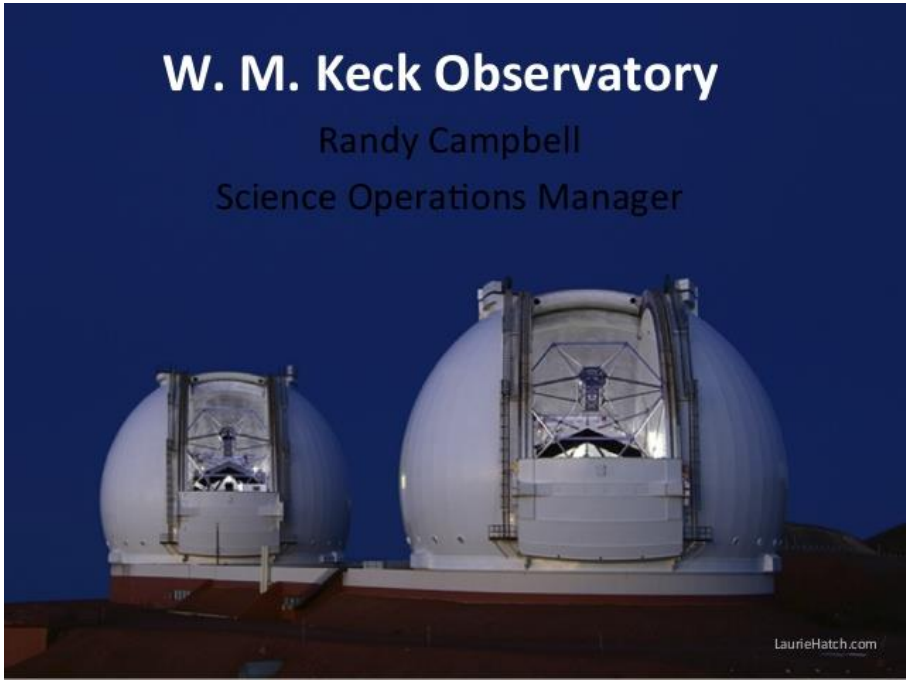
Course background and management

2022.3.8

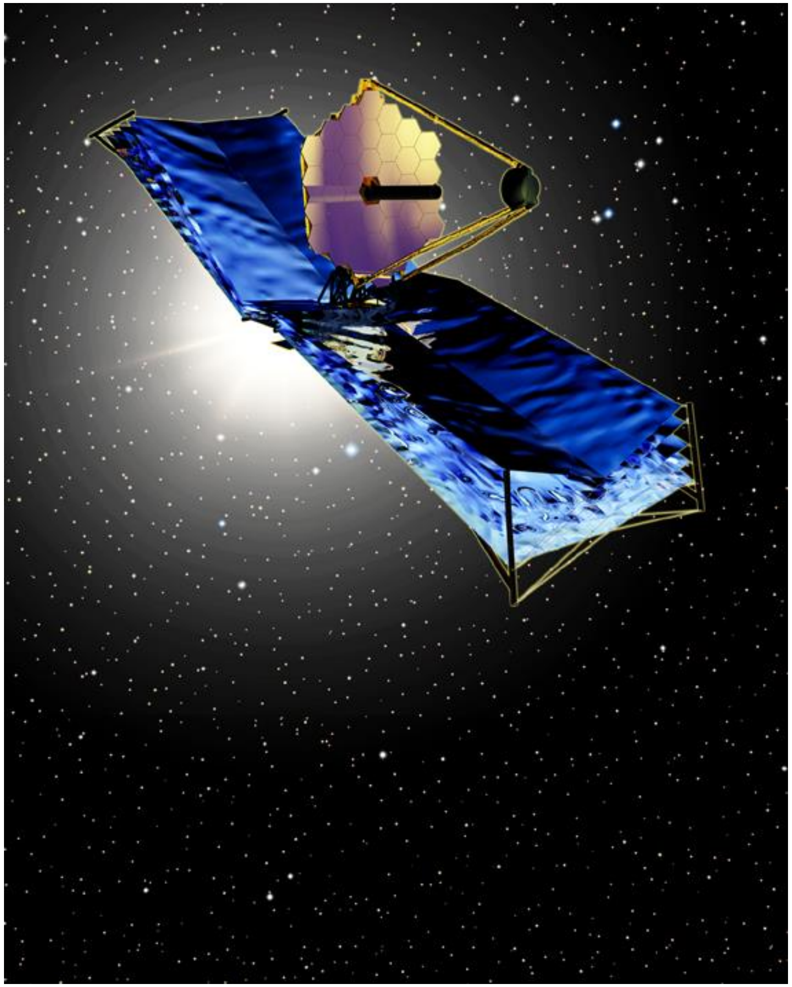
1610~ Galileo

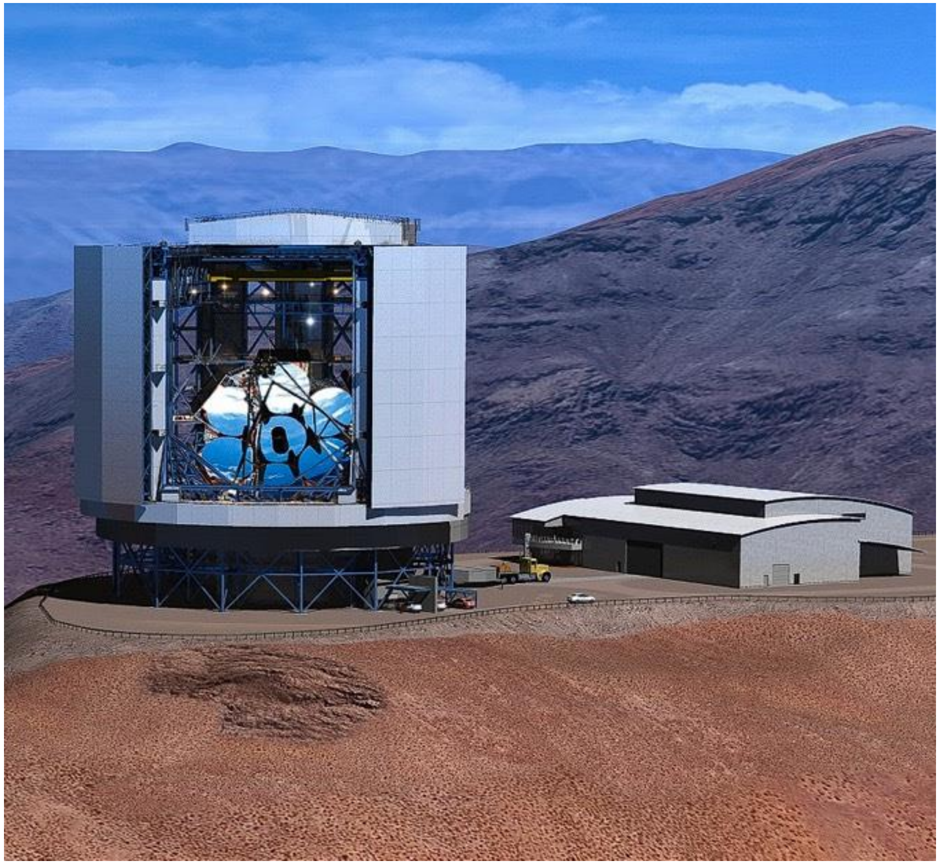


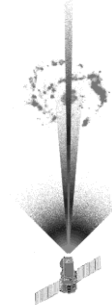
1990~


W. M. Keck Observatory
Randy Campbell
Science Operations Manager
LaurieHatch.com

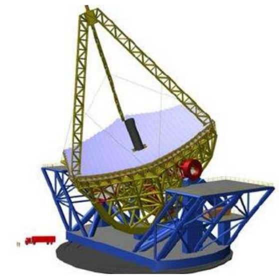
2020~

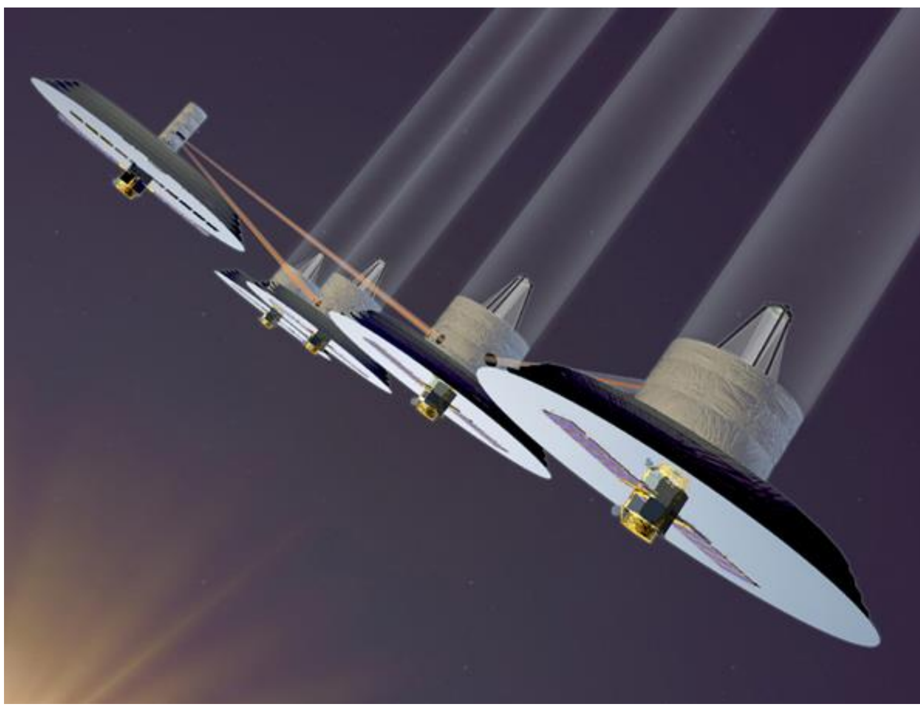






2040~

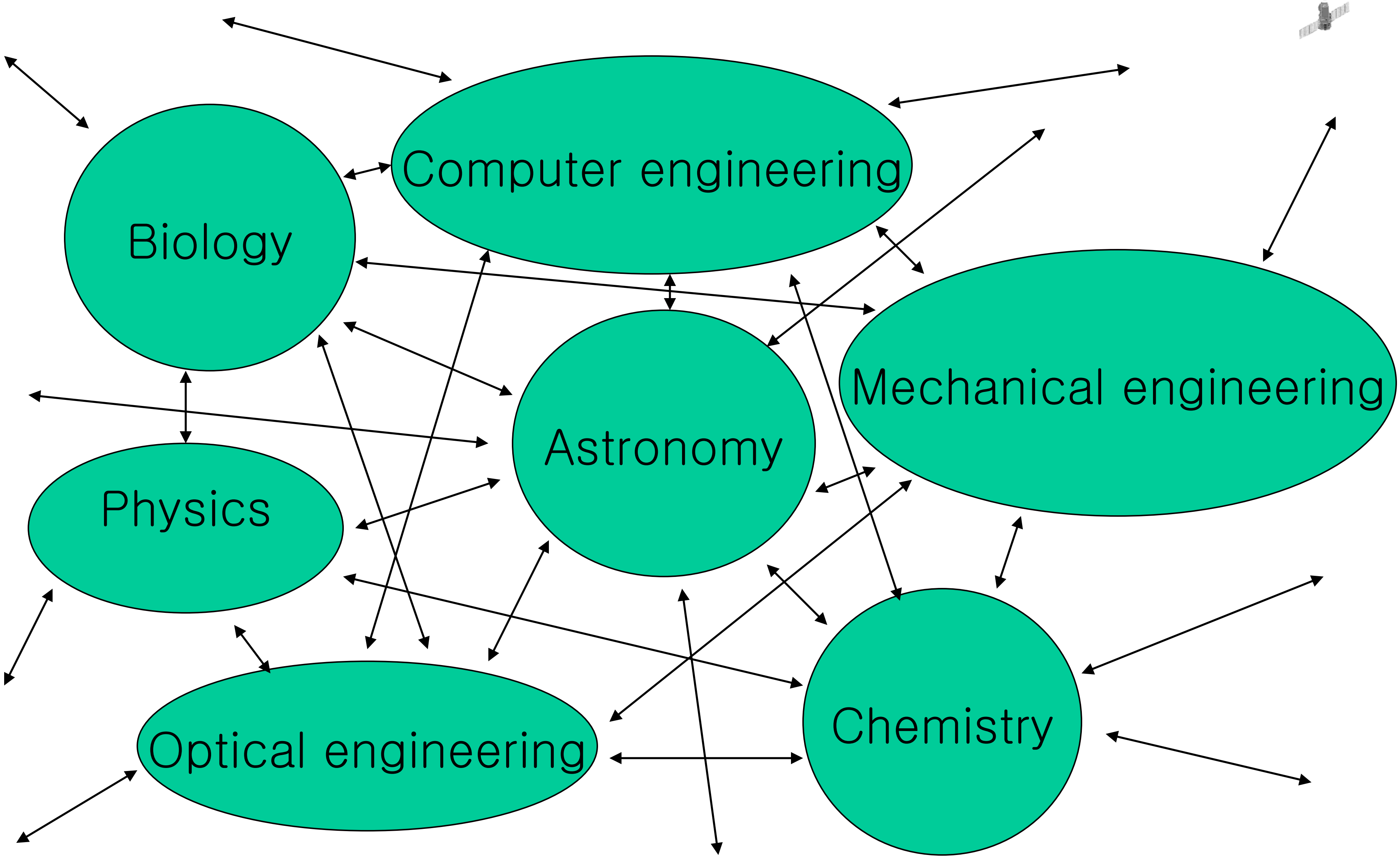




Philosophy

"…You learn only when you have done it…"

World of knowledge



Nature of knowledge

- Dead knowledge:
  - Facts, memorization, exact copy, no change, passive, listening, receipt, complete, perfect, boring
- Living knowledge:
  - Reasoning, application, logic, process, change, active, making, generating, incomplete, imperfect, exciting
- Useful only when
  - The “Dead” is converted to the “Living”
  - One has tools (methodology, training) in mind for initiation of such conversion whenever and wherever necessary.

World of knolwedge

- Conventional teaching-learning method: Newtonian paradigm, absolute framework, fixed shape
  - Dead knowledge, text book, well-defined, boundary
- Education revolution in teaching-learning methodology: Fluid paradigm, relative framework, changing shape
  - Living knowledge, frontier research, ill-defined, no boundary

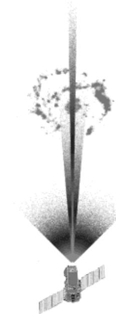
Two worlds: Old and New

Frontier knowledge
Unexplored
No absolute reference
No boundary
ill-defined
Uncomfortable, difficult
Research
New understanding
New Methods
New discovery
New education

Core knowledge
Well-defined
Text books
Existing understanding
Conventional methods
Explored
Comfortable, Easy
No new discovery
Boundary of idea imposed
Conventional education

Painful transition inevitable !

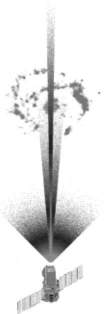
Where does the reality stand ?



- Astronomy and space science are fast-moving and dynamically changing disciplines.
  - NASA's HST, JWST, TPF, etc
- All other academic and industrial disciplines have the same nature too !
- Multi-disciplinary approaches are essential.
- The mind of explorer is necessary.

Lessons to be learnt

- Text books and existing research papers:
  - “stepping stones to boldly go where no man has ever gone before”
- Projects management:
  - “Process management”
- Collaboration:
  - “Collective generation of new knowledge”



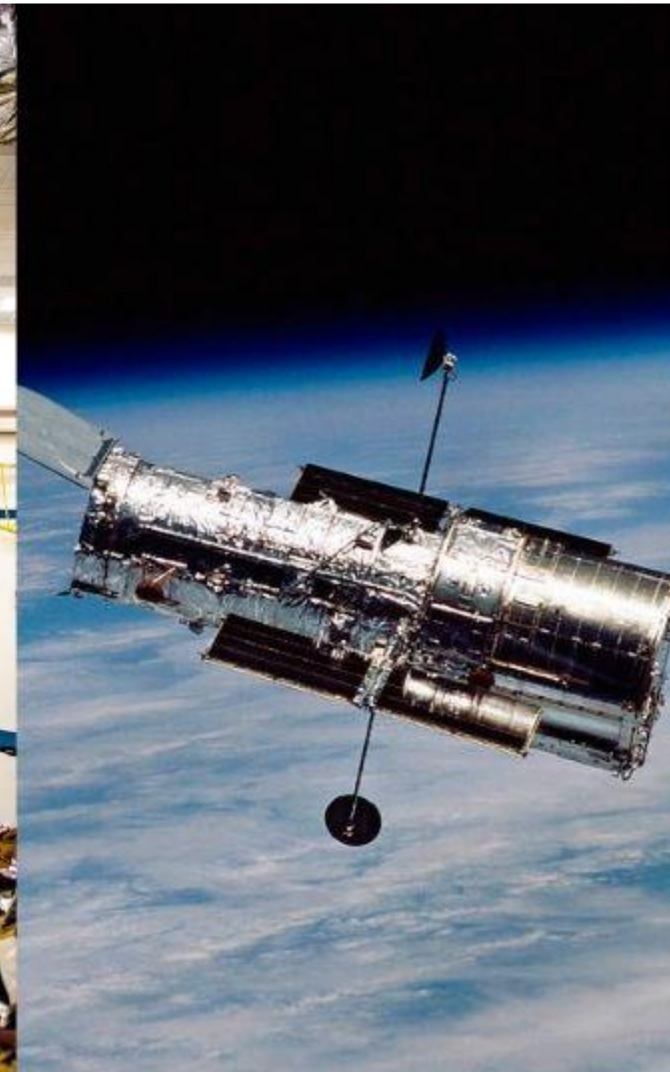
Course elements

- Phase 1: Studying text materials (5 weeks)
- Phase 2: Project definition (2 weeks)
- Phase 3: Student projects for (9 weeks)
- Academic conference (Dec)
- No exams, but project management and control

Grading scheme

- 30% : conference
- 30% : text material study
- 40% : weekly project review
- No mid term and final exams.

Text materials



Study methods for the 1st 5 weeks (Phase 1)

- Each student study and summarize
  - HST material for the first two weeks
  - JWST mayerial for the following three weeks

Roles and Responsibility for Phase 1

- Each student to summarize about 10~15 pages per week.
  - Present summary in class room
  - Submit ppt summary until midnight

Phase 2 study
2 weeks

- Students plan their space mission for astronomy for 2 weeks
  - Student project consists of about 100 WBS items
  - Those 100 WBSs are to be grouped to "Science Objectives", "Measurement Method", "Instruments", "Spacecraft, orbit and mission analysis" and "Expected results"

Phase 3 Study
9 weeks

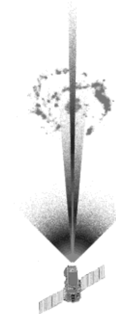
- Each student to apply the text material study to the selected project of their own
- Weekly review in quantitative project management
- Each student is to collect the study results of about 100 WBS items for the selected space mission

Academic conference

- Each student is to reorganize the Phase 3 study results to Conceptual Design Study material for space observation mission.
- Serves as “term end exam”

3^{rd} classroom hour assignment (March 8^{th})

- https://www.youtube.com/watch?v=lN1KJ8LYW3U
- Title: “Cosmic Journeys – Hubble: Universe in Motion”
- Submit the summary till midnight 8^{th} March 2022 (today)



March 15th Classroom

- Study and summarize HST material in 15 ppt pages: "Science Objectives", "Measurement method" and "Instruments"
- Submit the ppt file to kimsfield@gmail.com by midnight 12th March 2002.
- These materials to be presented to the classroom on March 15th